WMMSE for FR3 Coexistence in the GTA: A Study Using 3GPP/ITU-R Models and ISED Fixed-Service Data

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***Abstract*—This paper provides a practical and reproducible coexistence baseline for the frequency range 3 (FR3) downlink based on constrained weighted minimum mean-square error (WMMSE) beamforming. A key challenge in FR3 band is coexistence with incumbent terrestrial fixed service (FS) microwave links, whose protection is often specified through long-term interference-to-noise limits. We use a simplified wideband model that preserves full-band power and FS interference levels. The cellular channel model follows 3GPP TR 38.901 (UMi), while the BS-to-FS coupling uses ITU-R P.1411 above-rooftop propagation. To improve practical relevance, FS receiver locations and parameters are taken from the Innovation, Science and Economic Development Canada (ISED) Spectrum Management System (SMS) database for the Greater Toronto Area (GTA). Simulations are implemented in a TensorFlow/Sionna framework and compare four modes: No-FS, hard-null, budget-dual, and hybrid. Results show that No-FS, as expected, violates FS constraints by several orders of magnitude, while hard nulling enforces protection at a substantial sum-rate cost. In contrast, dual-based protection (budget-dual and hybrid) keeps the worst-case normalized interference close to the FS limit while achieving sum-rates close to the unconstrained solution.**

I. Introduction

The upper mid-band, often called "frequency range 3" (FR3), is attracting strong interest for 5G-Advanced and 6G. FR3 can offer much wider contiguous bandwidth than legacy sub-7 GHz deployments, while avoiding some of the severe coverage limits of millimeter-wave systems [1]–[3]. A key challenge is that FR3 is not an empty band, as several incumbents already operate in this frequency range. Therefore, practical cellular access in FR3 requires reliable coexistence mechanisms that control interference not only through power, but also directional transmission and spatial coordination [4], [5].

A major incumbent category is the terrestrial fixed service (FS), which includes point-to-point microwave links used for transport and access infrastructure [6]. FS protection is often expressed using long-term interference-to-noise (I/N) criteria. For example, ITU-R coordination guidance in nearby bands uses $I/N = -10\,\text{dB}$ as a long-term benchmark for compatibility evaluation [7]; recent simulation studies adopt similar metrics to assess feasibility under realistic deployments [8]. These observations motivate coexistence baselines that are transparent, reproducible, and aligned with incumbent protection metrics.

In this paper, we develop a practical FR3 coexistence baseline based on a constrained weighted minimum mean-square error (WMMSE) beamforming algorithm [9], [10]. Our goal is to provide a repeatable reference point for future coexistence strategies under realistic assumptions. To support this, we build a reproducible simulation framework using Sionna [11] for link and channel generation, follow standardized 3GPP channel modeling guidance (TR 38.901) [12], and use ITU-R propagation guidance relevant to short-range outdoor and over-rooftop conditions [13]. Importantly, we use real FS registrations from the Innovation, Science and Economic Development Canada (ISED) Spectrum Management System (SMS) database and evaluate coexistence for FS receivers in the Greater Toronto Area (GTA). We believe the use of real-world data significantly enhances the usefulness of this study.

Our main contributions are:

- A wideband OFDM system model with full-band power while accounting for interference.
- A tractable FS interference model that supports aggregation across base stations and bandwidth and maps naturally to long-term FS I/N protection criteria [7].
- A constrained WMMSE beamforming procedure with clear constraint handling and stopping rules, enabling reproducible comparisons and serving as a baseline for future improvements.

II. System Model

A. Network Topology and Wideband Abstraction

We model each sector as one base station (BS) transmitter. Let the set of BSs be $\mathcal{B} \triangleq \{1, \ldots, B\}$, where BS b has M transmit antennas. Each BS serves K user equipments (UEs) with one stream per UE. The UE set is $\mathcal{U} \triangleq \{1, \ldots, U\}$ with $U = BK$. Each UE has N_{r} receive antennas and is served by exactly one BS. We also consider L protected FS receivers indexed by $\mathcal{L} \triangleq \{1, \ldots, L\}$, where each FS receiver has a single antenna and is associated with an allocated center frequency and bandwidth.

Let $\mathbf{p}_b^{\text{BS}} \in \mathbb{R}^3$, $\mathbf{p}_u^{\text{UE}} \in \mathbb{R}^3$, and $\mathbf{p}_\ell^{\text{FS}} \in \mathbb{R}^3$ denote the BS, UE, and FS receiver locations, respectively. These locations determine BS–UE large-scale attenuation and BS–FS coupling, and they also define the BS-to-FS departure directions used in the steering-based FS model.

For notational convenience, we index the k-th UE served by BS b by the global index

$$u(b,k) \triangleq (b-1)K + k, \quad b \in \mathcal{B},\ k \in \{1,\dots,K\}. \quad (1)$$

We model a wide FR3 carrier by grouping the N subcarriers into T contiguous tone groups and scale per–group quantities by $\eta = N/T$ to preserve full-band power and interference.

B. Downlink Signal Model

On each tone group $t \in \mathcal{T}$, BS b transmits

$$\mathbf{x}_b[t] = \mathbf{W}_b[t]\,\mathbf{s}_b[t] \ \in \mathbb{C}^M, \quad (2)$$

where $\mathbf{W}_b[t] = \big[\mathbf{w}_{b,1}[t],\dots,\mathbf{w}_{b,K}[t]\big] \in \mathbb{C}^{M\times K}$ contains the linear beamformers and $\mathbf{s}_b[t] \in \mathbb{C}^K$ stacks the intended data symbols for the K users. We assume $\mathbb{E}\{\mathbf{s}_b[t]\mathbf{s}_b[t]^{\mathrm{H}}\} = \mathbf{I}_K$ and independent symbols across BSs.

The received vector at UE $u \in \mathcal{U}$ on tone group t is

$$\mathbf{y}_u[t] = \sum_{b\in\mathcal{B}} \mathbf{H}_{b,u}[t]\mathbf{x}_b[t] + \mathbf{n}_u[t], \quad (3)$$

where $\mathbf{H}_{b,u}[t] \in \mathbb{C}^{N_{\mathrm{r}}\times M}$ is the downlink channel from BS b to UE u, and $\mathbf{n}_u[t] \in \mathbb{C}^{N_{\mathrm{r}}}$ is additive noise.

Each UE applies a linear combiner $\mathbf{v}_u[t] \in \mathbb{C}^{N_{\mathrm{r}}}$ to detect its desired stream, resulting in $\hat{s}_u[t] = \mathbf{v}_u[t]^{\mathrm{H}}\mathbf{y}_u[t]$. To compactly represent inter-cell and intra-cell interference, we define the per-UE effective channel matrix

$$\mathbf{E}_u[t] \triangleq \big[\mathbf{H}_{1,u}[t]\mathbf{W}_1[t],\dots,\mathbf{H}_{B,u}[t]\mathbf{W}_B[t]\big] \in \mathbb{C}^{N_{\mathrm{r}}\times KB}. \quad (4)$$

By the indexing in (1), the desired stream for UE u corresponds to the u-th column of $\mathbf{E}_u[t]$, denoted by $\mathbf{e}_{u,u}[t] \in \mathbb{C}^{N_{\mathrm{r}}}$. We assume per-antenna circularly symmetric complex Gaussian noise with variance $\sigma^2_{u,\mathrm{eff}}(\gamma_{\mathrm{dB}}) = \sigma^2_u\, 10^{-\gamma_{\mathrm{dB}}/10}$.

III. Propagation and Fixed-Service Modeling

We now describe how the BS–UE channels are generated and how BS transmissions are mapped to long-term FS interference metrics.

A. BS–UE Channel Generation

We generate BS–UE large-scale attenuation using the 3GPP TR 38.901 urban micro (UMi) street-canyon model, including line-of-sight (LoS)/non-line-of-sight (NLoS) probability, distance-dependent pathloss, and log-normal shadow fading. For indoor UEs, we include an outdoor-to-indoor (O2I) penetration loss. In dB, the total large-scale loss is modeled as

$$\mathrm{PL}_{b,u} = \mathrm{PL}^{\mathrm{UMi}}_{b,u} + X_{b,u} + L^{\mathrm{O2I}}_u, \quad (5)$$

where $X_{b,u}$ is shadow fading and $L^{\mathrm{O2I}}_u = 0$ if O2I is disabled.

Given $\mathrm{PL}_{b,u}$, the linear gain is $\beta_{b,u} = 10^{-\mathrm{PL}_{b,u}/10}$. The small-scale fading is modeled as i.i.d. Rayleigh, and the downlink channel is

$$\mathbf{H}_{b,u}[t] = \sqrt{\beta_{b,u}}\,\tilde{\mathbf{H}}_{b,u}[t], \quad [\tilde{\mathbf{H}}_{b,u}[t]]_{i,j} \sim \mathcal{CN}(0,1), \quad (6)$$

for all $t \in \mathcal{T}$. Unless stated otherwise, we reuse the same large-scale gain across tone groups and treat the modeled band as frequency-flat at the tone-group level.

We next define the BS–FS coupling and FS protection thresholds used in the coexistence constraints.

B. Fixed-Service Statistics for Protection Constraints

FS protection is commonly defined over long averaging windows. For this reason, we enforce FS constraints using large-scale BS–FS coupling rather than instantaneous small-scale FS fading. Recall that $\mathcal{L}$ denotes the set of protected FS receivers, each assumed to be single-antenna.

a) Average BS–FS coupling.: We introduce an average large-scale gain $\bar{\beta}_{b,\ell}$ for each BS–FS pair (b,ℓ). This term captures the selected BS-to-FS pathloss model (in our simulations: ITU-R P.1411 above-rooftop ABG) as

$$\bar{\beta}_{b,\ell} = 10^{-\mathrm{PL}^{\mathrm{FS}}_{b,\ell}/10}\, G_\ell(b), \quad (7)$$

where $\mathrm{PL}^{\mathrm{FS}}_{b,\ell}$ is the BS–FS pathloss in dB, and $G_\ell(b)$ is the effective FS receiver gain towards BS b. When FS antenna metadata is available, $G_\ell(b)$ includes directional attenuation based on the angular mismatch. Otherwise the FS receiver is treated as omnidirectional.

b) Spatial coupling models.: For tone group t, define the BS transmit covariance

$$\mathbf{Q}_b[t] \triangleq \mathbf{W}_b[t]\mathbf{W}_b[t]^{\mathrm{H}} \succeq \mathbf{0}. \quad (8)$$

We consider two spatial coupling models as

$$\phi_{b,\ell}[t] \triangleq \begin{cases} \mathrm{tr}(\mathbf{Q}_b[t]), & \text{identity model,} \\ \mathbf{a}^{\mathrm{H}}_{b,\ell}\mathbf{Q}_b[t]\mathbf{a}_{b,\ell}, & \text{steering-rank-1 model,} \end{cases} \quad (9)$$

where $\mathbf{a}_{b,\ell} \in \mathbb{C}^M$ is a unit-norm steering vector (scaled such that $\|\mathbf{a}_{b,\ell}\|_2^2 = M$) constructed from BS array geometry and the BS-to-FS direction. The steering-rank-1 model captures the dependence of FS interference on BS transmit directionality, which is later needed for our *Hard-null* baseline.

c) Spectral overlap.: FS receiver ℓ occupies a narrow band around its center frequency $f^{\mathrm{FS}}_{c,\ell}$ with bandwidth B^{FS}_ℓ. Let $[f^{\min}_t, f^{\max}_t]$ denote the frequency span of tone group t, and let $[f^{\min}_\ell, f^{\max}_\ell]$ denote the FS band edges. The overlap fraction $\epsilon_{t,\ell} \in [0,1]$ is

$$\epsilon_{t,\ell} \triangleq \frac{\max\big(0,\ \min(f^{\max}_t, f^{\max}_\ell) - \max(f^{\min}_t, f^{\min}_\ell)\big)}{f^{\max}_t - f^{\min}_t}. \quad (10)$$

d) FS interference thresholds.: We represent FS protection using a per-receiver threshold $I_{\max,\ell}$ (Watts). When FS noise parameters are available, we compute the FS noise power N_ℓ over B^{FS}_ℓ and apply an interference-to-noise (I/N) target $\gamma^{\mathrm{IN}}_\ell$ (dB):

$$I_{\max,\ell} = N_\ell\, 10^{\gamma^{\mathrm{IN}}_\ell/10}. \quad (11)$$

These FS statistics, together with the MU-MIMO signal model, define the constrained beamforming problem stated next.

IV. Problem Formulation

In this section we formulate the corresponding wideband downlink beamforming problem. This formulation also clarifies the performance and protection metrics reported in Section VI.

A. Weighted Sum-Rate Objective

We consider a single scheduling instant and design linear precoders $\{\mathbf{W}_b[t]\}$ to maximize the average downlink sum-rate across the simulated tone groups. Let $\mathrm{SINR}_u[t]$ denote the post-combining signal-to-interference-plus-noise ratio (SINR) of UE u on tone group t under linear reception. The weighted sum-rate is given by

$$R(\{\mathbf{W}_b[t]\}) = \frac{1}{T}\sum_{t\in\mathcal{T}}\sum_{u\in\mathcal{U}} \xi_u \log_2\big(1+\mathrm{SINR}_u[t]\big), \tag{12}$$

where $\xi_u \geq 0$ are user weights (uniform by default).

B. Power and Fixed-Service Constraints

The simulator enforces a per-BS total transmit-power constraint across the full physical band. For BS b, the per-tone-group power is $\mathrm{tr}(\mathbf{Q}_b[t]) = \|\mathbf{W}_b[t]\|_{\mathrm{F}}^2$, and the wideband-scaled power is

$$P_b \triangleq \eta \sum_{t\in\mathcal{T}} \mathrm{tr}(\mathbf{Q}_b[t]). \tag{13}$$

The per-BS constraint is

$$P_b \leq P_{\mathrm{tot}}, \quad \forall b \in \mathcal{B}, \tag{14}$$

where P_{tot} is the BS power budget in Watts.

To protect FS receivers, we constrain the long-term average interference aggregated across BSs and tone groups. Using $\bar{\beta}_{b,\ell}$, $\epsilon_{t,\ell}$, and $\phi_{b,\ell}[t]$ from Section III-B, the interference at FS receiver ℓ is

$$I_\ell \triangleq \eta \sum_{t\in\mathcal{T}}\sum_{b\in\mathcal{B}} \epsilon_{t,\ell}\,\bar{\beta}_{b,\ell}\,\phi_{b,\ell}[t], \tag{15}$$

and the protection constraint is

$$I_\ell \leq I_{\max,\ell}, \quad \forall \ell \in \mathcal{L}. \tag{16}$$

In Section VI, we report the normalized worst-case interference $\max_{\ell\in\mathcal{L}} I_\ell / I_{\max,\ell}$ to quantify constraint satisfaction.

The optimization problem is thus to maximize (12) over $\{\mathbf{W}_b[t]\}$ subject to (14) and (16). The next section describes the constrained WMMSE baseline used to obtain stationary solutions of this non-convex problem.

V. WMMSE with Fixed-Service Protection

The constrained problem in Section IV is non-convex because SINRs are coupled across BSs and users. We therefore use a weighted minimum mean-square error (WMMSE) algorithm that alternates between MMSE receiver updates and beamformer updates, while enforcing the power and FS constraints through dual variables. This baseline defines the modes compared in simulations (No-FS, budget-dual, hard-null, and hybrid).

A. MMSE Receiver and MSE–SINR Relationship

We first introduce the MMSE quantities that link the sum-rate objective to a quadratic form. For each UE u and tone group t, received covariance (signal + interference + noise)

$$\mathbf{R}_u[t] \triangleq \mathbf{E}_u[t]\mathbf{E}_u[t]^{\mathrm{H}} + \sigma_{u,\mathrm{eff}}^2 \mathbf{I}_{N_{\mathrm{r}}}. \tag{17}$$

The linear MMSE combiner is

$$\mathbf{v}_u[t] = \mathbf{R}_u[t]^{-1}\mathbf{e}_{u,u}[t]. \tag{18}$$

The resulting minimum mean-square error (MSE) for estimating $s_u[t]$ is

$$e_u[t] = 1 - \Re\big\{\mathbf{e}_{u,u}[t]^{\mathrm{H}}\mathbf{v}_u[t]\big\}, \tag{19}$$

and the corresponding SINR is obtained as

$$\mathrm{SINR}_u[t] = \frac{1}{e_u[t]} - 1. \tag{20}$$

These expressions enable the standard WMMSE reformulation used for iterative beamformer updates as shown below:

B. WMMSE Weights and Beamformer Update

The WMMSE method alternates between updating (i) the MMSE combiners $\{\mathbf{v}_u[t]\}$, (ii) the MSE weights $\{q_u[t]\}$, and (iii) the transmit beamformers $\{\mathbf{W}_b[t]\}$. Steps (i) are given by (18)–(19). The weights are updated as

$$q_u[t] = \frac{\xi_u}{e_u[t]}, \quad \forall u \in \mathcal{U},\ t \in \mathcal{T}. \tag{21}$$

To express the beamformer update compactly, define the back-projected vector

$$\mathbf{g}_{b,u}[t] \triangleq \mathbf{H}_{b,u}[t]^{\mathrm{H}}\mathbf{v}_u[t] \in \mathbb{C}^M. \tag{22}$$

For each (b,t), form

$$\mathbf{D}_b[t] = \sum_{u\in\mathcal{U}} q_u[t]\,\mathbf{g}_{b,u}[t]\mathbf{g}_{b,u}[t]^{\mathrm{H}} \in \mathbb{C}^{M\times M}, \tag{23}$$

and the right-hand side collecting only the K UEs served by BS b:

$$\mathbf{C}_b[t] = \sum_{k=1}^{K} q_{u(b,k)}[t]\,\mathbf{g}_{b,u(b,k)}[t]\,\mathbf{e}_k^{\top} \in \mathbb{C}^{M\times K}, \tag{24}$$

where $\mathbf{e}_k$ is the k-th canonical basis vector in $\mathbb{R}^K$.

To enforce the constraints, we introduce nonnegative dual variables μ_b for the BS power constraints (14) and λ_ℓ for the FS constraints (16). With a small parameter $\rho > 0$ for numerical stability, the beamformer update is

$$\mathbf{W}_b[t] = \Big(\mathbf{D}_b[t] + (\eta\mu_b + \rho)\mathbf{I}_M + \mathbf{F}_b[t]\Big)^{-1}\mathbf{C}_b[t], \tag{25}$$

where $\mathbf{F}_b[t]$ is the FS penalty term. For the identity FS model,

$$\mathbf{F}_b[t] = \Big(\eta \sum_{\ell\in\mathcal{L}} \lambda_\ell\ \epsilon_{t,\ell}\,\bar{\beta}_{b,\ell}\Big)\mathbf{I}_M, \tag{26}$$

and for the steering-rank-1 model,

$$\mathbf{F}_b[t] = \eta \sum_{\ell\in\mathcal{L}} \lambda_\ell\ \epsilon_{t,\ell}\,\bar{\beta}_{b,\ell}\,\mathbf{a}_{b,\ell}\mathbf{a}_{b,\ell}^{\mathrm{H}}. \tag{27}$$

We next summarize how the dual variables are updated and how this leads to the coexistence modes compared in the simulations.

C. Dual Updates and Coexistence Modes

After each primal update, we ensure feasibility of the BS power constraint using a per-BS scaling (projection) applied uniformly across tone groups:

$$\mathbf{W}_b[t] \leftarrow \alpha_b \, \mathbf{W}_b[t], \quad \alpha_b = \min\left(1, \sqrt{\frac{P_{\text{tot}}}{P_b}}\right), \tag{28}$$

where P_b is given in (13).

The dual variables are updated by projected gradient ascent with fixed step sizes ρ_μ and ρ_λ:

$$\mu_b \leftarrow \left[\mu_b + \rho_\mu\big(P_b - P_{\text{tot}}\big)\right]_+, \quad \forall b \in \mathcal{B}, \tag{29}$$

$$\lambda_\ell \leftarrow \left[\lambda_\ell + \rho_\lambda\big(I_\ell - I_{\max,\ell}\big)\right]_+, \quad \forall \ell \in \mathcal{L}, \tag{30}$$

where I_ℓ is given in (15).

1) FS enforcement modes: The following modes are used in Section VI:

- *No-FS:* disable FS protection by setting $\lambda_\ell = 0$ for all ℓ.
- *Budget-dual:* enforce (16) via the dual update (30).
- *Hard-null (steering-rank-1):* in addition to the updates above, project $\mathbf{W}_b[t]$ onto the nullspace of selected FS steering vectors. Let $\mathcal{L}_b \subseteq \mathcal{L}$ denote the set of FS receivers whose null directions are enforced at BS b, and define $\mathbf{A}_b \triangleq \left[\mathbf{a}_{b,\ell_1}, \ldots, \mathbf{a}_{b,\ell_{|\mathcal{L}_b|}}\right] \in \mathbb{C}^{M\times|\mathcal{L}_b|}$. The projection is

$$\mathbf{W}_b[t] \leftarrow \left(\mathbf{I}_M - \mathbf{A}_b\big(\mathbf{A}_b^{\text{H}}\mathbf{A}_b\big)^{\dagger}\mathbf{A}_b^{\text{H}}\right)\mathbf{W}_b[t], \tag{31}$$

  where $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse. This mode is intentionally conservative and aims to drive FS interference close to zero.
- *Hybrid:* apply the budget-dual mode, and fall back to the hard-null projection when the FS constraints remain violated under the given iteration budget.

D. Stopping Criterion

The WMMSE iterations stop when the relative change of the beamformers becomes small or when a maximum iteration budget is reached:

$$\frac{\left\|\mathbf{w}^{(k+1)} - \mathbf{w}^{(k)}\right\|_2}{\left\|\mathbf{w}^{(k)}\right\|_2} \le \varepsilon, \qquad \mathbf{w}^{(k)} \triangleq \operatorname{vec}\big(\{\mathbf{W}_b[t]\}_{b,t}\big), \tag{32}$$

where $\varepsilon > 0$ is a tolerance and $\operatorname{vec}(\cdot)$ stacks all beamforming coefficients.

VI. Simulation Results

This section evaluates the proposed FR3 coexistence framework in an urban deployment with real FS incumbents. We first describe the simulation setup and the considered methods; we then report on the rate–protection trade-off, solver convergence, user-rate distribution, and an SNR sensitivity sweep.

A. Simulation Setup

We use a reproducible TensorFlow-based simulator built on Sionna, with 3GPP-compliant stochastic channel and signal generation. The cellular system operates at carrier frequency $f_c = 8.2$ GHz over total bandwidth $B_{\text{tot}} = 200$ MHz with subcarrier spacing $\Delta f = 120$ kHz. Each sector is subject to a total transmit power constraint of $P_{\max} = 49$ dBm across the full band, $N = 1667$ subcarriers. User thermal noise is modeled with temperature $T_{\text{N}} = 290$ K and noise figure $\text{NF} = 9$ dB, which yields a baseline noise power of approximately $\sigma_u^2 \approx -114.2$ dBm per resource element (RE).

The cellular deployment follows an urban microcell (UMi) setting with 61 tri-sector sites ($B = 183$ sectors). Each sector is equipped with $M = 32$ base-station (BS) antennas and serves 10 user equipments (UEs), each with $N_r = 4$ receive antennas. Users are uniformly dropped within a disk of radius $R_{\text{UE}} = 100$ m around their serving sector. We model a high indoor load with indoor probability 0.8. Indoor propagation follows the 3GPP outdoor-to-indoor (O2I) low-/high-loss model, with 80% low-loss and 20% high-loss buildings, and indoor penetration distance sampled uniformly in $[0, 25]$ m. Large-scale cellular attenuation follows the 3GPP TR 38.901 UMi street-canyon model with log-normal shadow fading of 4.0 dB (LoS) and 7.82 dB (NLoS).

Remark: Throughout the simulations, the reported *sum-rate* (in bps/Hz) refers to the *network* sum spectral efficiency, i.e., $\sum_{u\in\mathcal{U}} \log_2\big(1 + \text{SINR}_u\big)$, rather than a per-user value (simply obtained by dividing by $|\mathcal{U}| = 1830$).

Incumbent protection is based on real FS registrations obtained from the ISED SMS database, filtered to the GTA. To study an extreme yet realistic coexistence regime, we generate cellular sites using a clustered layout in which FS receiver locations act as hotspots: 75% of the 61 sites are clustered around FS receivers, and the remaining 25% are deployed uniformly as a spatial background over the GTA footprint.

Fig. 1 shows the simulated GTA footprint with the cellular site locations and the registered FS receivers. The bottom panel is a zoomed-in view of downtown Toronto, illustrating the local density of sites and user drops relative to nearby incumbents.

The FS receiver front-end is modeled with bandwidth 30 MHz and noise figure 5 dB. The incumbent protection constraint is expressed via an interference-to-noise (I/N) target of -10 dB, yielding an interference threshold $I_{\max} = N_{\text{FS}} + \text{I/N}$ unless explicitly overridden. Large-scale BS-to-FS attenuation follows the ITU-R P.1411 above-rooftop alpha–beta–gamma (ABG) model with UMi-based LoS probability.

We compare four modes: *No-FS* (no incumbent constraint), *Hard-null* (conservative nulling toward FS), *Budget-dual* (dual-based interference budgeting), and *Hybrid* (combined strategy).

B. Trade-off Versus FS I/N Target

Fig. 2 shows how each method balances cellular performance and FS protection as the I/N target changes, *evaluated at* $\text{SNR} = 0$ dB. In the top panel, *Hard-null* is consistently the

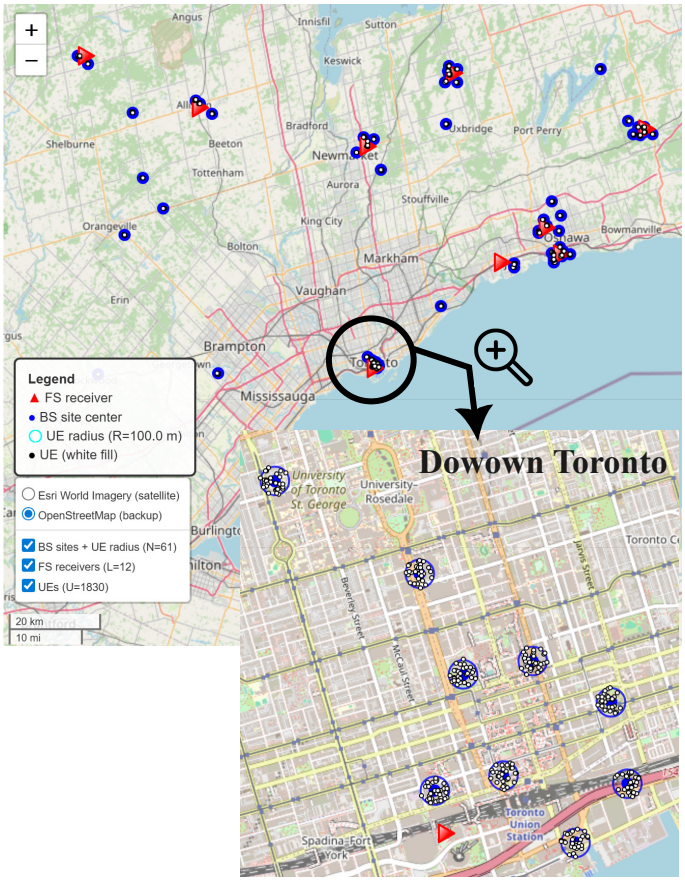

Fig. 1. GTA simulation layout showing cellular sites and incumbent FS receiver locations; the bottom panel is a zoomed-in view of downtown Toronto.

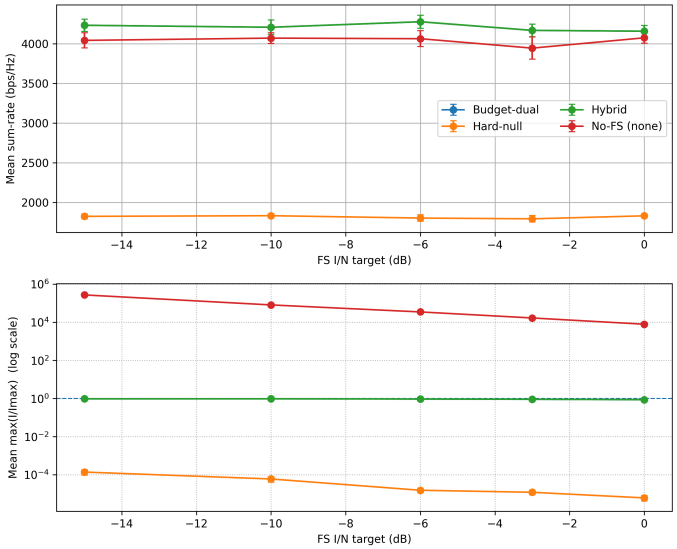

Fig. 2. Mean *network sum-rate* (top; sum over $U = 1830$ UEs, in bps/Hz) and mean normalized FS interference $\max(I/I_{\max})$ (bottom) versus FS I/N target.

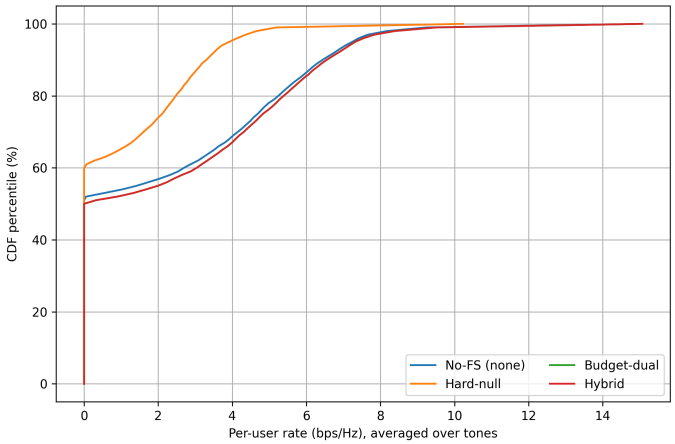

Fig. 3. CDF of *per-user* spectral efficiency (in bps/Hz, averaged over tones) under different coexistence methods.

lowest-rate solution (a network sum-rate of ≈ 1.78–1.89 kbps/Hz in this experiment), while *Budget-dual*, *Hybrid*, and *No-FS* yield a network sum-rate 4–4.3 kbps/Hz.

The key difference appears in the bottom panel, which reports the mean of $\max(I/I_{\max})$, where I is the aggregate BS→FS interference power and $I_{\max}$ is its I/N-based threshold. As expected, *No-FS* produces very large normalized interference values (on the order of 10^3–10^6 across the sweep), indicating violations of the FS constraint by several orders of magnitude. In contrast, *Budget-dual* and *Hybrid* keep $\max(I/I_{\max})$ close to one (typically between 0.6 and 1.0), which indicates that they operate near the protection boundary and use the interference budget efficiently. *Hard-null* drives $\max(I/I_{\max})$ close to zero, which reflects an excessive conservative design that sacrifices spatial degrees of freedom and reduces sum-rate.

In this configuration, *Budget-dual* and *Hybrid* overlap almost perfectly across the sweep. This suggests that, under the current parameters, the hybrid mechanism provides limited additional benefit beyond the budget-dual design. One detail in Fig. 2 is that *No-FS* is not always the highest mean sum-rate; in some cases, *Budget-dual* or *Hybrid* slightly outperform it. This does not contradict the optimization problem, since the unconstrained feasible set contains the constrained one. An explanation is algorithmic: the reported values come from a non-convex iterative solver with a fixed iteration budget, and different modes may converge to different stationary points.

C. User-Rate Distribution

Fig. 3 complements the mean-rate analysis by showing the empirical cumulative distribution function (CDF) of user rates (evaluated at SNR= 0 dB and $I/N = -10$ dB). The *Hard-null* curve is shifted to the left, indicating that a larger fraction of users experience lower rates, which is consistent with strong nulling reducing beamforming gain across the population. In contrast, *Budget-dual* and *Hybrid* almost overlap with *No-FS*, suggesting that the dual-based coexistence mechanisms preserve the overall rate distribution while enforcing FS protection. Overall, the dual-based designs reduce leakage while maintaining user-level performance.

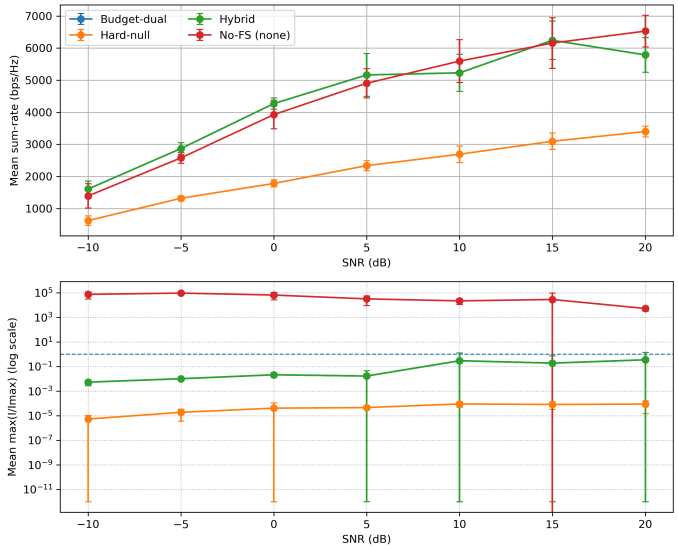

Fig. 4. Mean *network sum-rate* (top; sum over $U = 1830$ UEs, in $\mathrm{bps/Hz}$) and mean normalized FS interference $\max(I/I_{\max})$ (bottom) versus UE SNR.

A notable feature is the sharp CDF jump at $0\ \mathrm{bps/Hz}$, indicating that a non-negligible fraction of UEs achieve near-zero instantaneous rate. Such behavior is expected under an instantaneous weighted sum-rate objective with uniform weights and no explicit fairness or minimum-rate constraints: the WMMSE stationary point may assign negligible beamforming gain to very weak users, effectively dropping them in that realization. The effect is amplified in our FR3 setting ($f_c = 8.2$ GHz) by the heavy indoor load (0.8) and the TR 38.901 outdoor-to-indoor (O2I) low-/high-loss penetration model with indoor depth up to $25\ \mathrm{m}$, which yields a substantial population of deep-indoor/high-loss links. Hard-null further consumes spatial degrees of freedom, increasing the near-zero-rate mass and shifting the CDF left.

D. Sensitivity to UE SNR

Fig. 4 reports mean network sum-rate and mean $\max(I/I_{\max})$ versus SNR and serves as a sensitivity check (I/N fixed at -10 dB). The top panel shows the expected increase in sum-rate with SNR for all modes, with *Hard-null* remaining separated by a large gap. At higher SNR, *No-FS* becomes the best-performing scheme, which matches the intuition that relaxing FS constraints provides more optimization freedom when UE noise is low.

The bottom panel highlights that coexistence constraints remain essential: *No-FS* violates the FS budget by four to five orders of magnitude across almost the entire SNR range (the ratio decreases from about 10^5 to $\sim 10^3$ at 20 dB, but it remains far above the threshold). The protected methods remain close to the $\max(I/I_{\max}) \approx 1$ regime. The decreasing trend of *No-FS* with increasing SNR is consistent with the simulator implementation, where SNR is realized via UE noise scaling rather than increased BS transmit power. Overall, the sweep confirms that unconstrained leakage can remain orders of magnitude above the FS threshold even when UE SNR is high.

VII. Conclusion

We presented a reproducible FR3 coexistence baseline for multi-cell downlink MU-MIMO with explicit FS protection constraints, combining a simplified wideband model with standardized propagation (3GPP TR 38.901 for BS–UE and ITU-R P.1411 for BS–FS) and *real FS registrations from the ISED SMS database in the GTA*. Our results show a clear protection–throughput trade-off: No-FS achieves high sum-rate but violates FS limits by orders of magnitude, hard nulling is overly conservative with large rate loss, and the dual-based design operates close to the FS boundary while keeping user-rate performance close to the unconstrained case. In our setup, the hybrid mode largely overlaps with the budget-dual solution, indicating that dual-based protection is sufficient for the tested I/N targets and SNR values, and providing a practical reference baseline for FR3 coexistence studies.

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